New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: _____ SIGNATURE: _____

ID NUMBER: _____ EXAM ROOM: M A T H _____ GROUP: _____

Practice Final Exam

COURSE NAME: Computer Programming I COURSE NUMBER: CS 102

EXAMINATION DURATION: 90 MINUTES EXAM VERSION Practice

ADDITIONAL MATERIALS ALLOWED TO USE: No additional materials or items are allowed to be used.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- Read and strictly follow the extra instructions at the beginning of each Section.
- For Section A, mark your answers **only on the OMR answer sheet**; answers written in the booklet are not graded.
- Put your signature and write your ID on every page.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Section & Type of Questions	Questions	Points	Score
Section A: Multiple Choice	1–20	40	
Section B: Fill in the Gaps	21–26	30	
Section C: Code Writing	27–29	30	
Total		**100**	

Name :	
Exam :	CS102 Computer Programming I — Final
Date :	

Roll No

0	○	○	○	○	○	○
1	○	○	○	○	○	○
2	○	○	○	○	○	○
3	○	○	○	○	○	○
4	○	○	○	○	○	○
5	○	○	○	○	○	○
6	○	○	○	○	○	○
7	○	○	○	○	○	○
8	○	○	○	○	○	○
9	○	○	○	○	○	○

Computer Programming I

Section A: MCQ

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○

	A	B	C	D
11	○	○	○	○
12	○	○	○	○
13	○	○	○	○
14	○	○	○	○
15	○	○	○	○
16	○	○	○	○
17	○	○	○	○
18	○	○	○	○
19	○	○	○	○
20	○	○	○	○

Use a **black or blue ink pen**. Fill one circle per question completely. This sheet will be scanned electronically — keep it clean.

SIGNATURE: ID:

Section A: Multiple-Choice Questions (40 points)

Instructions: Each question is worth 2 points. Fill the corresponding circle of the correct answer on the OMR sheet given on the previous page. **Only the OMR sheet will be evaluated.** If the OMR sheet is left empty, no score will be awarded regardless of answers written in the booklet.

1. (2 points) The binary number 11010_2 equals which decimal value?
   A. 24
   B. 26
   C. 28
   D. 22

2. (2 points) How many *bits* are there in one byte?
   A. 4
   B. 16
   C. 8
   D. 32

3. (2 points) In a flowchart, an *oval* symbol represents:
   A. A process
   B. A decision
   C. Input
   D. Start / End

4. (2 points) Which of the following is *NOT* a valid C++ identifier?
   A. `3rd`
   B. `total`
   C. `_x`
   D. `name1`

5. (2 points) What is the value of `19 % 4`?
   A. 4
   B. 3
   C. 4.75
   D. 1

6. (2 points) In C++, the expression `8 / 3` (both operands `int`) evaluates to:
   A. 2.67
   B. 3
   C. 2
   D. 2.5

7. (2 points) What is the value of `(4 < 2) || (6 > 1)`?
   A. false
   B. 0
   C. error
   D. true

8. (2 points) How many times does the body run: `for(int i=0; i<4; i++)`?

A. 4
B. 3
C. 5
D. 0

9. (2 points) Which keyword *immediately exits* a loop?
   A. `continue`
   B. `break`
   C. `return`
   D. `exit`

10. (2 points) The line declaring a function *before* `main` (its interface) is called the:
   A. definition
   B. call
   C. prototype
   D. body

11. (2 points) A function that calls itself is called:
   A. iterative
   B. overloaded
   C. inline
   D. recursive

12. (2 points) What does the standard `factorial(5)` return?
   A. 120
   B. 25
   C. 60
   D. 720

13. (2 points) After `int a[4] = {1, 2};` the value of `a[3]` is:
   A. 2
   B. 0
   C. garbage
   D. error

14. (2 points) Which sorting method repeatedly swaps *adjacent* out-of-order elements?
   A. Binary
   B. Linear search
   C. Bubble sort
   D. Hashing

15. (2 points) For `string s = "hello";`, what is `s.length()`?
   A. 4
   B. 6
   C. 0
   D. 5

16. (2 points) How many elements does `int m[2][3]` have?
   A. 6

B. 5
C. 23
D. 3

17. (2 points) The operator `&x` gives:
    A. the value of x
    B. the address of x
    C. a copy of x
    D. nothing

18. (2 points) The expression `new int` returns:
    A. an integer value
    B. an array
    C. a pointer (address)
    D. nothing

19. (2 points) Which stream type is used to *write* to a file?
    A. `ifstream`
    B. `cin`
    C. `istream`
    D. `ofstream`

20. (2 points) A `struct` is used to group:
    A. related data of possibly different types
    B. only integers
    C. only functions
    D. loops

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Section B: Fill in the Gaps (30 points)

Each blank is worth 5 marks. Choose the correct operator or value from the Word Bank and write it in the box. Read carefully — some items differ by a single character. Not all words are used; each word is used at most once.

WORD BANK

`&&` `||` `<` `<=` `==` `++` `--` `%` `/` `'\0'`

21. (5 points) To test that `x` is between 1 and 10 *inclusive* (both conditions must hold), write `x>=1` _____ `x<=10`.

22. (5 points) To visit array indices `0` through `n-1` exactly, the loop header is `for(int i=0; i` _____ `n; i++)`.

23. (5 points) To test whether two values are *equal* inside an `if`, use the _____ operator.

24. (5 points) To compute the *remainder* of `a` divided by `b`, write `a` _____ `b`.

25. (5 points) The operator that adds exactly 1 to a variable `i` is `i`_____.

26. (5 points) A `char` array used as a C-string must have, as its final stored element, the value _____.

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Section C: Code Writing (30 points)

Write clear, correct C++. You only need the requested function (and any helper), not a whole program unless asked. Each problem is worth 10 marks.

27. (10 points) Write a function `int countEven(int a[], int n)` that returns how many elements of the array `a` of size `n` are **even**.

28. (10 points) Write a **recursive** function `int factorial(int n)` that returns $n!$ for $n \geq 0$ (recall $0! = 1$).

29. (10 points) A student is described by `struct Student { string name; double gpa; };`. Given an array `Student cls[]` of `n` students ($n \geq 1$), write code (a function or a fragment) that finds and **prints the name of the student with the highest GPA**.

END OF PRACTICE EXAMINATION